INCH-POUND
MIL-C-55302/117D(USAF)
6 MAY 1993
SUPERSEDING
MIL-C-55302/117C(USAF)
21 September 1988

MILITARY SPECIFICATION SHEET

CONNECTORS, PRINTED CIRCUIT SUBASSEMBLY AND ACCESSORIES
PLUG, RIGHT ANGLE, DECADE INCREMENT 20 THROUGH 110 AND 128 CONTACT POSITIONS, FOR PRINTED WIRING BOARDS (.050 SPACING)

This specification sheet is approved for use by Rome Air Development Center, Department of the Air Force, and is available for use by all Departments and Agencies of the Department of Defense

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation MIL-C-55302

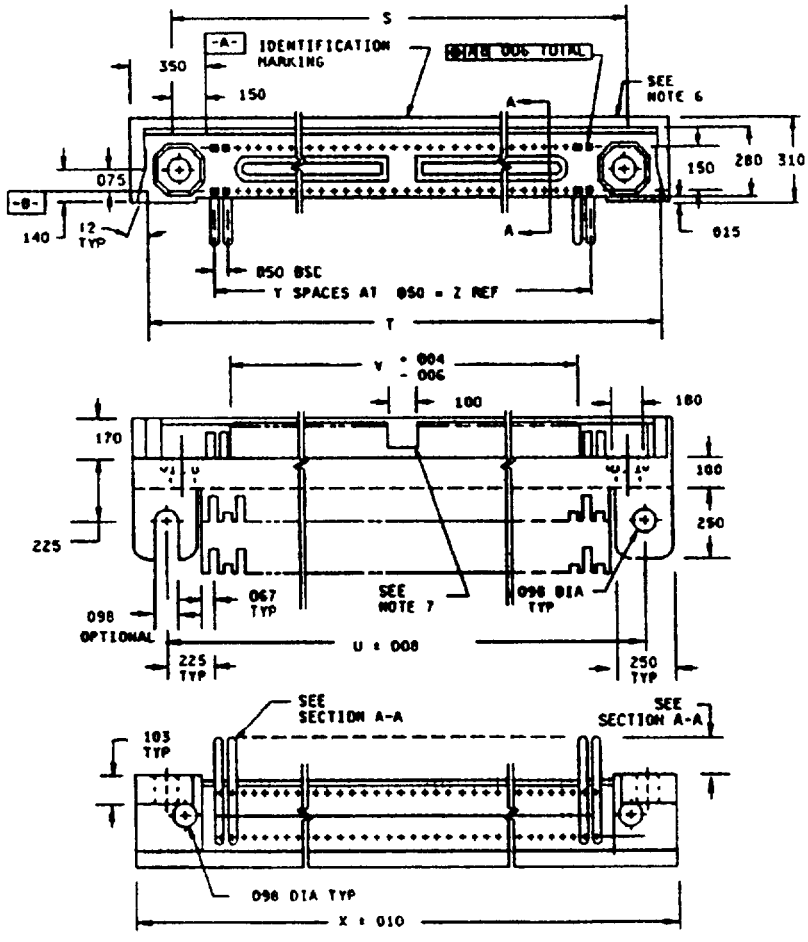
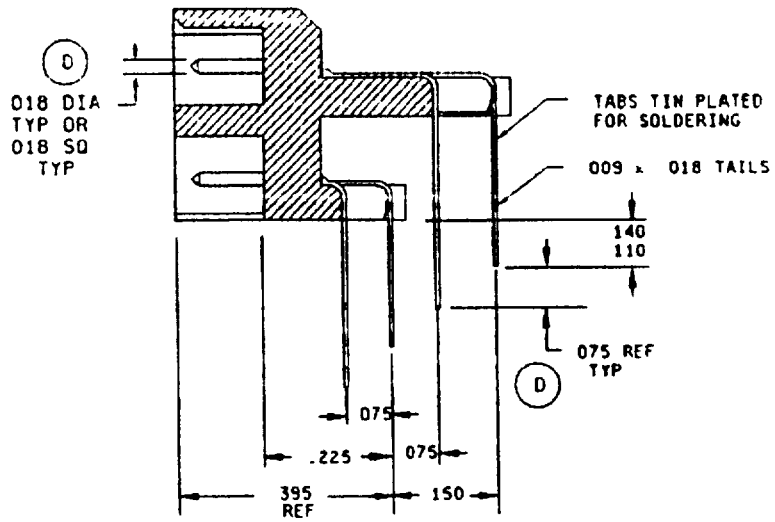


FIGURE 1 Connectors, plug (.050 (1.27 mm) square grid)

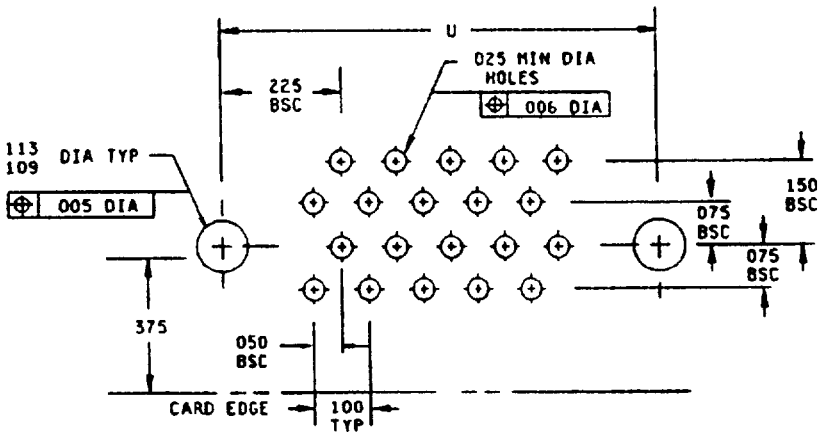
(D) denotes changes

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SECTION A-A



RECOMMENDED PRINTED WIRING BOARD LAYOUT

Inches	mm	Inches	mm	Inches	mm
004	0 10	075	1 90	170	4 32
005	0 13	098	2 49	180	4 57
006	0 15	.100	2 54	225	5 72
008	0 20	103	2 62	250	6 35
.009	0.23	109	2.77	.280	7 11
010	0 25	110	2 79	310	7 87
018	0 46	113	2 87	350	8 89
050	1 27	140	3 56	375	9 52
067	1 70	150	3 81	395	10 03

FIGURE 1 Connector, plug .050 (1.27 mm) spacing - Continued

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NOTES
1 Dimensions are inches
2 Metric equivalents are given for general information only
3 Unless otherwise specified, tolerance is ±.005 (0 13 mm) on decimals
4 These connectors mate with connectors specified in MIL-C-55302/119(USAF).
5 Pads suitable for printed wiring board support are optional
6 Numbers indicating end positions stamped this side only
7 Void is not present on 20 through 50 position

FIGURE 1 Connector, plug .050 (1 27 mm) spacing - Continued

TABLE I. Dash numbers and dimensions

Dash number	No of contacts	Dimensions							Contact identification numbers
		S	T	U	V	X	Y	Z	
01	20	.750 (19.50)	.970 (24.64)	900 (22.86)	290 (7.37)	1.150 (29.21)	9	.450 (11.43)	1-10
02	30	1.000 (25.40)	1 220 (30.99)	1 150 (29.21)	.540 (13.72)	1.400 (35.56)	14	.700 (17.78)	1-15
03	40	1.250 (31 75)	1 470 (37 34)	1 400 (35.56)	790 (20 07)	1.650 (41 91)	19	.950 (24 13)	1-20
04	50	1 500 (38.10)	1 720 (43.69)	1.650 (41.91)	1 040 (26.42)	1.900 (48 26)	24	1 200 (30 48)	1-25
05	60	1 750 (44.45)	1.970 (50.04)	1 900 (48.26)	1.290 (32 77)	2 150 (54.61)	29	1 450 (36.83)	1-30
06	70	2.000 (50.80)	2.220 (56.39)	2 150 (54.61)	1.540 (39 12)	2 400 (60.96)	34	1.700 (43 18)	1-35
07	80	2.250 (57.15)	2.470 (62.74)	2.400 (60.96)	1.790 (45 47)	2.650 (67.31)	39	1.950 (49.53)	1-40
08	90	2.500 (63.50)	2.720 (69.09)	2.650 (67.31)	2.040 (51 82)	2.900 (73.66)	44	2.200 (55 88)	1-45
09	100	2.750 (69.85)	2 970 (75.44)	2.900 (73.66)	2.290 (58.17)	3.150 (80.01)	49	2 450 (62.23)	1-50
10	110	3.000 (76.20)	3 220 (81.79)	3.150 (80.01)	2 540 (64.52)	3 400 (86.36)	54	2.700 (68 58)	1-55
11	128	3 450 (87 63)	3 670 (93.22)	3 600 (91 44)	2 990 (75 95)	3 850 (97 79)	63	3 150 (80.01)	1-64

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REQUIREMENTS

Design and construction

Dimensions and configurations See figure 1

Material

(D) Insulator body Insulator material shall be in accordance with MIL-C-55302 or MIL-M-24519 type GLCP-30F

(D) Contact material Contact material shall be in accordance with MIL-C-55302 or beryllium copper alloy CDA174 in accordance with ASTM B768

(D) Plating Contact plating shall be gold in accordance with MIL-C-55302 over nickel in accordance with MIL-C-55302 on the contact engagement area for 100 inch minimum length The termination area plating shall be tin lead in accordance with MIL-C-55302 over nickel in accordance with MIL-C-55302 to within 020 inches maximum of plastic housing body The remaining areas of the contact shall be nickel plated in accordance with MIL-C-55302

Contact identification: See figure 1

Contact rating: 1 5 amperes per contact, maximum

Operating temperature -65°C to +125°C

Keying See MIL-C-55302/31

Mating and unmating. The maximum mating force in pounds shall be the number of contacts multiplied by 30 and the withdrawal force in pounds shall be a minimum of 03 times the number of contacts and shall not exceed the measured insertion force

Contact resistance: The average contact resistance of all contacts measured shall not exceed 025 ohm, and no individual contact pair shall have a resistance exceeding 030 ohm

Contact retention 75 pound, minimum

Dielectric withstanding voltage

Sea level 600 volts rms

High altitude: 150 volts rms

Solderability Each connector shall be subjected to method 208 of MIL-STD-202 (solder termination only) (applicable to qualification and retention of qualification only)

Part number M55302/117 - (dash number from table I)

Patent number 3,404,367. The Government has a royalty free license under this patent for the benefit of manufacturers of the item either for the Government or for use in equipment to be delivered to the Government

CONCLUDING MATERIAL

Custodian
Air Force - 17

Review activities
Air Force - 11, 85, 99
DLA - ES

User activity
Air Force - 19

Preparing activity
Air Force - 17

Agent
DLA - ES

(Project 5935-F405-02)

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